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<https://swayam.gov.in>

https://swayam.gov.in/nc_details/NPTEL

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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 11: Assignment 11 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) Given n samples $x_1, x_2, \dots, x_N$ drawn independently from an Exponential distribution unknown parameter λ , find the MLE of λ .

1 point

$$\lambda_{MLE} = \sum_{i=1}^n x_i$$



$$\lambda_{MLE} = n \sum_{i=1}^n x_i$$



$$\lambda_{MLE} = \frac{n}{\sum_{i=1}^n x_i}$$



$$\lambda_{MLE} = \frac{\sum_{i=1}^n x_i}{n}$$



$$\lambda_{MLE} = \frac{n-1}{\sum_{i=1}^n x_i}$$



$$\lambda_{MLE} = \frac{\sum_{i=1}^n x_i}{n-1}$$

2) Suppose we are trying to model a p dimensional Gaussian distribution. What is the actual number of independent parameters that need to be estimated? **1 point**



2

Week 10 ()**Week 11 ()**

- ☐ Gaussian Mixture Models (unit? unit=122&less on=123)
- ☐ Expectation Maximization (unit? unit=122&less on=124)
- ☐ Expectation Maximization - Continued (unit? unit=122&less on=125)
- ☐ **Practice: Week 11: Assignment 11 (Non Graded) (assessment? name=274)**

- ☐ Week 11 Feedback Form : Introduction To Machine Learning (unit? unit=122&less on=292)

- ☒ Week 11: Solution (unit? unit=122&less on=230)

- ☒ Quiz: Week 11 : Assignment 11 (assessment? name=303)

Week 12 ()**Text Transcripts ()**☐ p ☐ $2p$ ☐ $p(p+1)$ ☐ $p(p+1)/2$ ☐ $p(p+3)/2$

3) Given n samples $x_1, x_2, \dots, x_N$ drawn independently from a Poisson distribution **1 point** unknown parameter λ , find the MLE of λ .

☐

$$\lambda_{MLE} = \sum_{i=1}^n x_i$$

☐

$$\lambda_{MLE} = n \sum_{i=1}^n x_i$$

☐

$$\lambda_{MLE} = \frac{n}{\sum_{i=1}^n x_i}$$

☐

$$\lambda_{MLE} = \frac{\sum_{i=1}^n x_i}{n}$$

☐

$$\lambda_{MLE} = \frac{n-1}{\sum_{i=1}^n x_i}$$

☐

$$\lambda_{MLE} = \frac{\sum_{i=1}^n x_i}{n-1}$$

4) In Gaussian Mixture Models, π_i are the mixing coefficients. Select the correct conditions that the mixing coefficients need to satisfy for a valid GMM model.

1 point☐

$$-1 \leq \pi_i \leq 1, \forall i$$

☐

$$0 \leq \pi_i \leq 1, \forall i$$

☐

$$\sum_i \pi_i = 1$$

☐

$$\sum_i \pi_i \text{ need not be bounded}$$

5) Expectation-Maximization, or the EM algorithm, consists of two steps - E step and the M-step. Using the following notation, select the correct set of equations used at each step of the algorithm.

1 point

Notation.

X Known/Given variables/data**Z** Hidden/Unknown variables θ Total set of parameters to be learned θ_k Values of all the parameters after stage k **Q**(,) The Q-function as described in the lectures

**Download
Videos ()**

Books ()

**Problem
Solving
Session -
July 2024 ()**

☐ $E - \mathbb{E}_{Z|X,\theta} [\log(\text{Pr}(X, Z|\theta_m))]$

☐ $E - \mathbb{E}_{Z|X,\theta_{m-1}} [\log(\text{Pr}(X, Z|\theta))]$

☐ $M - \underset{\theta}{\operatorname{argmax}} \sum_Z \text{Pr}(Z|X, \theta_{m-1}) \cdot \log(\text{Pr}(X, Z|\theta))$

☐ $M - \underset{\theta}{\operatorname{argmax}} Q(\theta, \theta_{m-2})$

☐ $M - \underset{\theta}{\operatorname{argmax}} Q(\theta, \theta_{m-1})$

Check Answers and Submit

